

10 MILES PER HOUR

THE AVERAGE SPEED OF DRAIS'S LAUFMASCHINE



Karl von Drais's Laufmaschine (running machine) was a forerunner of the bicycle and the first two-wheeled form of personal transportation.



each ring refracts light at a different angle

stepped lens focuses the beam

AUGUSTIN-JEAN FRESNEL (1788–1827)

Fresnel worked as an engineer during the Napoleonic Wars of 1803 to 1815. Afterward, he began to research light and optics, making key contributions to understanding the nature of light waves, diffraction, and polarization. He is best known for his invention of the stepped glass Fresnel lens, which is commonly used in lighthouses.



THE THEORY THAT LIGHT TRAVELS IN WAVES (see 1801) was backed up in 1816 by a series of precise experiments with diffraction—the way light spills around objects into shadows—by French engineer **Augustin-Jean Fresnel**. When Fresnel shone a light through slits, he detected tiny fringes of light that could only be produced by interference between waves. Fresnel backed this up with detailed calculations of **how a light wave might move and produce diffraction**.

Working with French physicist **François Arago** (1786–1853), in

Fresnel lighthouse lens
His experiments in light and optics led Fresnel to create a special lens for use in lighthouses; it was also sometimes used in theater lights. It focuses the light using multiply stepped glass, rather than a single thick lens.

1817 Fresnel began to explore polarization, which was then thought irreconcilable with wave theory. **Polarized light is reflected in just one plane**; Fresnel found that beams of

polarized light do not create interference fringes if they are polarized in different planes.

In 1816, British physicist **David Brewster** (1781–1868) **calculated Brewster's Angle**—the angle at which light must strike an object at for maximum polarization.

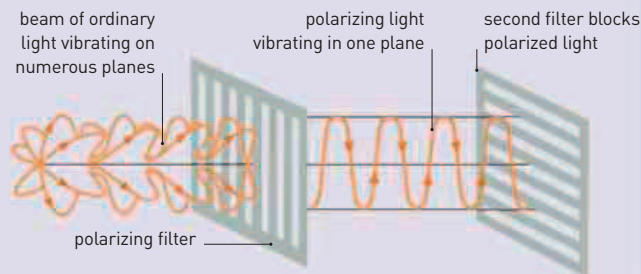
In 1817, **three new elements were discovered**: cadmium by German chemist **Friedrich Stromeyer** (1776–1835); lithium by Swedish chemist Johann Arfwedson (1792–1841); and selenium by Swedish chemist Jöns Berzelius (1779–1848).

In Germany, Baron **Karl von Drais** (1785–1851) **invented an early bicycle** known as the Laufmaschine—propelled by feet rather than pedals. His first public ride took place on June 12, 1817.

POLARIZATION

Augustin-Jean Fresnel worked out that light moves forward in waves that vibrate transversely—perpendicularly to the direction in which they are traveling.

Ordinary light vibrates at every angle or plane, but when it is polarized—by passing through a polarizing filter—the vibrations are reduced to a single plane.



1816 David Brewster calculates **Brewster's Angle** of polarization

1816 The first wire-cable suspension bridge, the Spider Bridge at Falls of Schuylkill, Philadelphia, opens

1816 British engineer Robert Stirling obtains a patent for the Stirling hot air engine

1816 Augustin-Jean Fresnel finds evidence for the **wave theory** of light in diffraction

1817 Augustin-Jean Fresnel and François Arago begin their experiments on polarization

1817 British physicist David Brewster patents the **kaleidoscope**

1817 Friedrich Stromeyer discovers **cadmium**

June 12, 1817 Baron Karl von Drais takes the first public ride on his **Laufmaschine**

1817 Johann Arfwedson discovers **lithium**

1817 Jöns Berzelius discovers **selenium**